

## Unit 402 Field Data Review

Mohawk Valley West Slope demonstration. **Cable** unit, 38.9 ac. 19 plots, BAF 20, variable-radius cruise on a 300 ft grid, 203 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

**TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE**

| STAND DENSITY        |                                        | STAND INFO          |                                                                                                                                       |
|----------------------|----------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Basal area           | 214 sq ft/ac (1 SE of mean, BA 3.1 %)  | Type                | Sierran mixed conifer                                                                                                                 |
| Trees per acre       | 64                                     | CWHR size / density | 5 / D (cover 90 %)                                                                                                                    |
| QMD                  | 24.8 in                                | SDI                 | 261 (45 % of max 576)                                                                                                                 |
| SAWTIMBER (>= 10 in) |                                        | BIOMASS (< 10 in)   |                                                                                                                                       |
| BA   QMD             | 214 sq ft/ac   24.8 in                 | BA   QMD            | 0 sq ft/ac   0.0 in                                                                                                                   |
| TPA                  | 64                                     | TPA                 | 0                                                                                                                                     |
| Volume               | 40.6 MBF/ac (8094 cu ft)               | Volume              | 0.0 green tons/ac (0 cu ft)                                                                                                           |
| TREATMENT TARGET     |                                        | CRUISE DESIGN       |                                                                                                                                       |
| Leave                | 35 % of max SDI, about 165 sq ft/ac BA | Vol. sampling error | 14.4 % at 95 % (CV 30 %, t = 2.10, n = 19)                                                                                            |
| Species (BA)         | PP 67, WF 65, DF 36, IC 27, SP 18      | Stratum standard    | 30 % of volume (R5 2409.12-2021-4, sec. 41.1): <b>MEETS</b> ; plots for standard 5                                                    |
|                      |                                        | Sale volume share   | 2.1 % of sale volume: standard shown for information (under the R5 10 % threshold)                                                    |
|                      |                                        | BA sampling error   | 6.6 % at 95 % (CV 14 %); reference, not the standard                                                                                  |
|                      |                                        | Measured trees      | PP 64, WF 62, DF 34, SP 17, IC 26; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): <b>ok</b> |
|                      |                                        | Plot count          | 19; R5 20-plot minimum for an area-based cruise (sec. 41.3): <b>SHORT</b> by 1                                                        |

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise\_data.xlsx, Standards and assumptions.

**TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE**

| DBH class, in | Trees      | TPA         | BA           | cu ft       | MBF         |
|---------------|------------|-------------|--------------|-------------|-------------|
| 12-15.9       | 14         | 13.1        | 14.7         | 360         | 1.8         |
| 16-19.9       | 30         | 17.5        | 31.6         | 872         | 4.4         |
| 20-23.9       | 33         | 13.0        | 34.7         | 1056        | 5.3         |
| 24-27.9       | 28         | 8.0         | 29.5         | 1017        | 5.1         |
| 28-31.9       | 17         | 3.6         | 17.9         | 659         | 3.3         |
| 32-35.9       | 12         | 2.0         | 12.6         | 534         | 2.7         |
| 36-39.9       | 20         | 2.7         | 21.1         | 900         | 4.5         |
| 40-43.9       | 10         | 1.1         | 10.5         | 500         | 2.5         |
| 44-47.9       | 15         | 1.4         | 15.8         | 811         | 4.1         |
| 48-51.9       | 5          | 0.4         | 5.3          | 269         | 1.3         |
| 52-55.9       | 3          | 0.2         | 3.2          | 171         | 0.9         |
| 56-59.9       | 4          | 0.2         | 4.2          | 241         | 1.2         |
| 60-63.9       | 12         | 0.6         | 12.6         | 706         | 3.5         |
| <b>All</b>    | <b>203</b> | <b>63.8</b> | <b>213.7</b> | <b>8094</b> | <b>40.6</b> |

**TABLE 3. STOCK TABLE BY SPECIES, PER ACRE**

| Species    | Trees      | TPA         | BA           | QMD         | cu ft       | MBF         |
|------------|------------|-------------|--------------|-------------|-------------|-------------|
| PP         | 64         | 16.8        | 67.4         | 27.2        | 2968        | 14.9        |
| WF         | 62         | 20.7        | 65.3         | 24.1        | 2377        | 11.9        |
| DF         | 34         | 12.0        | 35.8         | 23.4        | 1254        | 6.3         |
| IC         | 26         | 9.5         | 27.4         | 23.0        | 818         | 4.1         |
| SP         | 17         | 4.8         | 17.9         | 26.1        | 677         | 3.4         |
| <b>All</b> | <b>203</b> | <b>63.8</b> | <b>213.7</b> | <b>24.8</b> | <b>8094</b> | <b>40.6</b> |

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 402 plots, 300 ft grid (orange: QA flag)

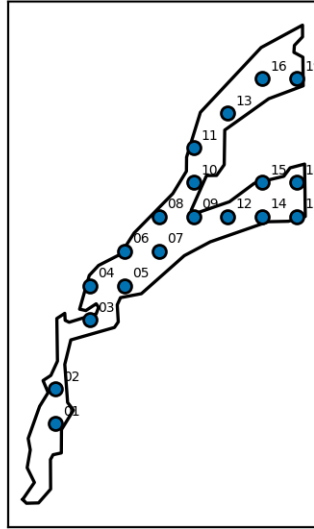


Figure 1. Plot locations with QA status.

**QA findings:** none; unit accepted; 1 more plots to reach the R5 20-plot minimum (sec. 41.3)

TABLE 4. PLOT LIST

| Plot   | Trees in | BA sq ft/ac | Slope % | Status |
|--------|----------|-------------|---------|--------|
| 402-01 | 7        | 140         | 37      | ok     |
| 402-02 | 12       | 240         | 53      | ok     |
| 402-03 | 10       | 200         | 44      | ok     |
| 402-04 | 12       | 240         | 61      | ok     |
| 402-05 | 12       | 240         | 83      | ok     |
| 402-06 | 9        | 180         | 36      | ok     |
| 402-07 | 11       | 220         | 48      | ok     |
| 402-08 | 10       | 200         | 56      | ok     |
| 402-09 | 10       | 200         | 34      | ok     |
| 402-10 | 12       | 240         | 58      | ok     |
| 402-11 | 9        | 180         | 39      | ok     |
| 402-12 | 11       | 220         | 76      | ok     |
| 402-13 | 10       | 200         | 61      | ok     |
| 402-14 | 11       | 220         | 62      | ok     |
| 402-15 | 10       | 200         | 42      | ok     |
| 402-16 | 12       | 240         | 57      | ok     |
| 402-17 | 12       | 240         | 46      | ok     |
| 402-18 | 13       | 260         | 51      | ok     |
| 402-19 | 10       | 200         | 57      | ok     |